HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Assisted with the **planning and design** of **port infrastructure**, contributing to preliminary assessments for a new quay wall structure.
- Generated **feasibility studies and business cases** for coastal resilience projects, incorporating **environmental considerations** and **sea level rise data**.
- Produced technical **reports, presentations, and calculations** for **waterfront development** projects, ensuring adherence to project specifications.
- Collaborated with a **multidisciplinary team** of engineers to develop practical guidance on **coastal erosion** mitigation strategies.
- Undertook **site visits and surveys** for an ongoing **quay wall upgrade** design, documenting observations and engaging with **project stakeholders**.
- Supported the development of **port masterplans** for enterprise clients, integrating data analysis from various sources to inform strategic decisions.
- Containerized **Java Spring Boot** microservices using **Docker** and managed deployments on **AWS EC2** instances, improving application portability.

SKILLS

Engineering Principles – Maritime Engineering, Civil Engineering, Coastal Engineering, Geotechnical Engineering, Structural Engineering, Naval Architecture, Floating Structures

Software & Tools – Python, Java, SQL, Spring Boot, Docker, AWS, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau, R, pandas, NumPy, scikit-learn, Flask, FastAPI

Data Analysis & BI – Data Analytics, Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, Statistical Analysis, Regression, Hypothesis Testing, Exploratory Data Analysis, A/B Testing, Time-Series Analysis, Machine Learning Fundamentals

Financial Systems – Payments Operations Dashboard, Transaction Monitoring, Reconciliation Workflows, ERP Integration Support, Change Management

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux Shell Scripting

Communication & Collaboration – Written and Verbal Communication, Collaborative Work, Stakeholder Engagement, Client Interaction, Teamwork

PROJECTS

Coastal Resilience Planning Tool (MSc Project)

- Developed a **Python-based tool** to model coastal erosion scenarios, integrating historical weather data and topographical surveys.
- Implemented **data visualization** using **pandas** and **Matplotlib** to present simulation outcomes, aiding in the assessment of resilience strategies.
- Generated comprehensive technical documentation and a final report outlining the tool's capabilities and potential applications for infrastructure planning.

Port Infrastructure Design Support System (HCL Technologies)

- Engineered **Python scripts** to automate the extraction and validation of design parameters from client specification documents for port facilities.
- Developed **SQL queries** against PostgreSQL to manage and retrieve design data for a portfolio of maritime infrastructure projects, supporting **design engineers**.
- Contributed to the development of **REST APIs** using **Java Spring Boot** for data exchange between design modules and operational monitoring systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)